Testleaf

Always Ahead

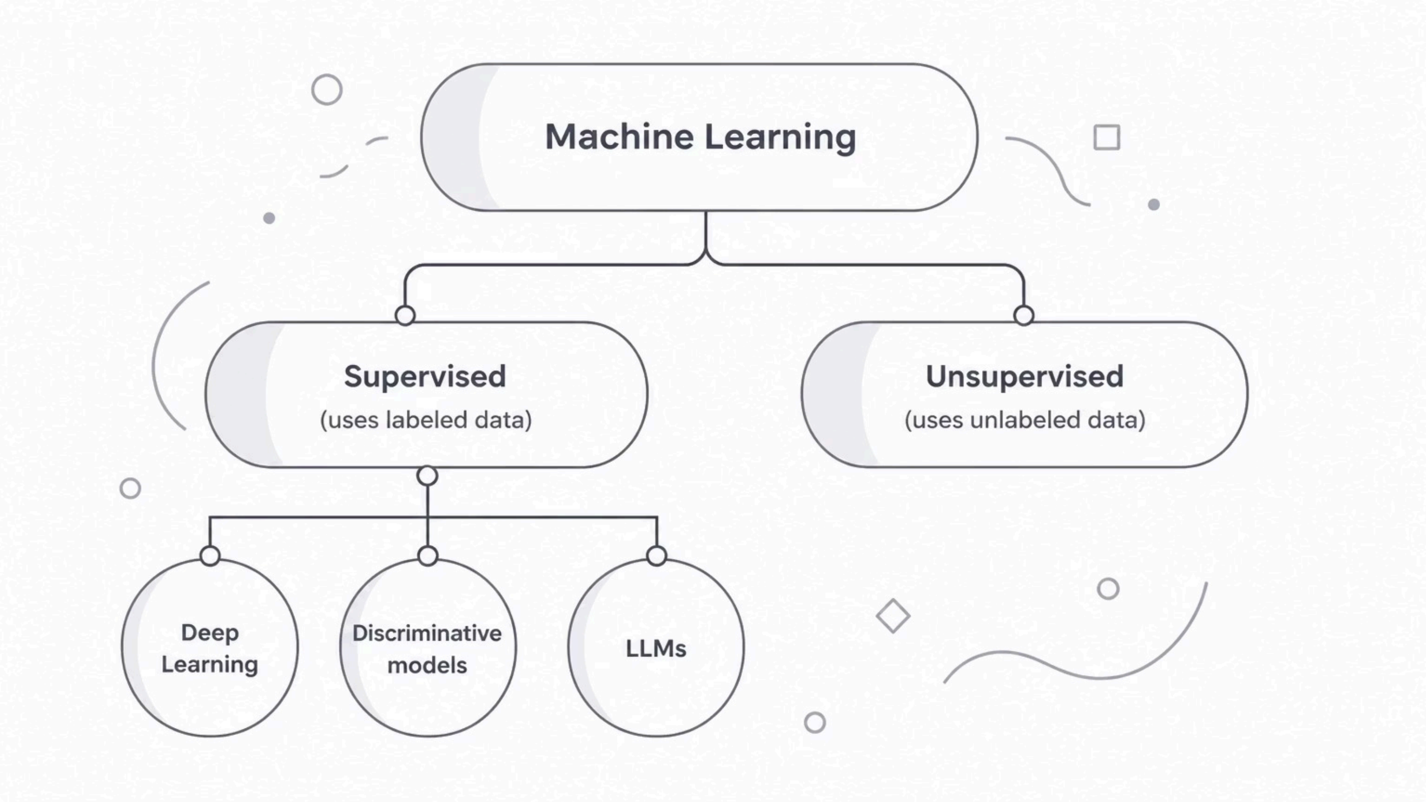
Machine Learning Types

A foundational overview of how machines learn — from supervised and unsupervised models to deep learning and large language models. This guide breaks down the core categories every student and practitioner should know.

Types of ML

OVERVIEW

Machine learning models fall into distinct families based on how they consume data and generate outputs. The two foundational branches — **supervised** and **unsupervised** learning — differ primarily in whether the training data carries labels. From there, specialized paradigms like deep learning, discriminative models, and large language models (LLMs) extend these foundations into powerful real-world applications.



Supervised models learn from labeled examples, mapping inputs to known outputs. Unsupervised models discover hidden patterns without any predefined labels — making them powerful for clustering and dimensionality reduction tasks.

Supervised Learning

CORE CONCEPT

Supervised learning is the most widely used paradigm in machine learning. The model is trained on a dataset where every input example is paired with a correct output label. Over many iterations, the algorithm learns to map inputs to outputs — enabling accurate predictions on new, unseen data.

Labeled Training Data

Every example in the dataset includes both input features and the correct answer, guiding the model during training.

Two Main Tasks

Regression predicts continuous values. **Classification** assigns inputs to discrete categories.

Generalization

The ultimate goal is a model that performs well on data it has never seen before, not just on training examples.

Linear Regression Algorithm

SUPERVISED LEARNING

What Is It?

A **regression algorithm** in supervised learning that **predicts numerical values** based on input features. Linear regression models the relationship between one or more input variables and a continuous output by fitting a straight line (or hyperplane) through the training data.

The model learns the best-fit parameters — slope and intercept — by minimizing the difference between predicted and actual values, typically using a technique called **least squares**.

Linear regression is often the first algorithm introduced in ML courses because it is interpretable, computationally efficient, and forms the conceptual foundation for more complex models.

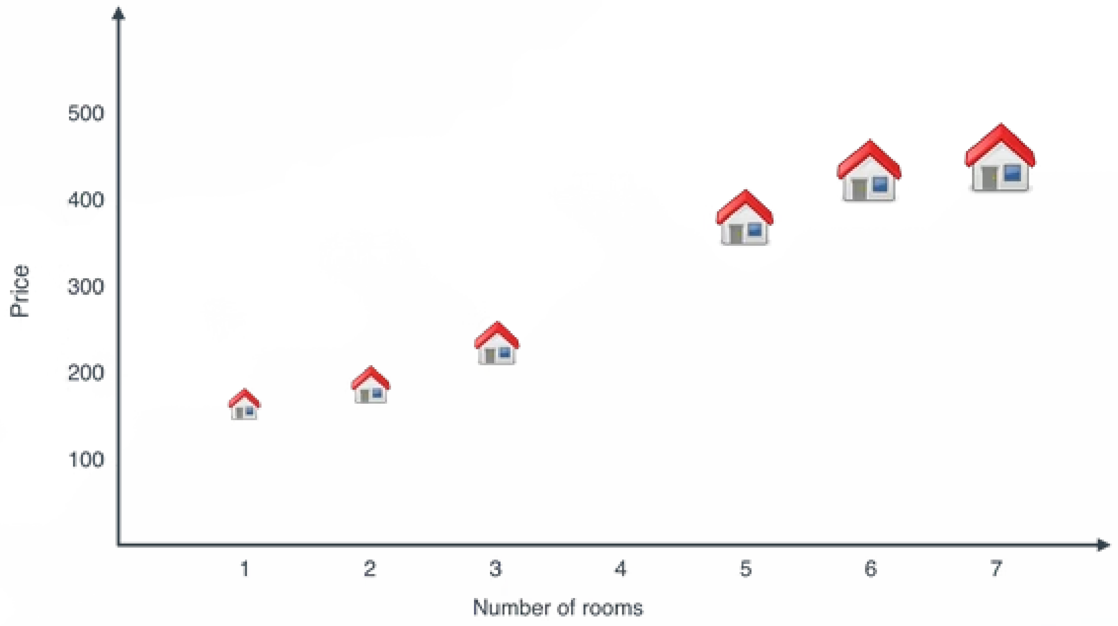
Key Characteristics

- Predicts a **continuous numeric output**
- Assumes a linear relationship between inputs and output
- Trained by minimizing prediction error
- Output is a real number, not a category
- Simple, fast, and highly interpretable

Linear Regression in Action

SUPERVISED LEARNING EXAMPLE

One of the most intuitive applications of linear regression is **house price prediction**. By learning from historical data — including features like number of rooms, location, and square footage — the model draws a line of best fit through the data points and uses it to estimate the price of a new home.



What the Chart Shows

As the number of rooms increases from 1 to 7, the predicted house price rises steadily — illustrating a clear **positive linear relationship** between input feature and output value.

Continuous Value Prediction

Unlike classification, which outputs a category, linear regression outputs a **real number** — such as $375,000 — making it ideal for pricing, forecasting, and estimation tasks.